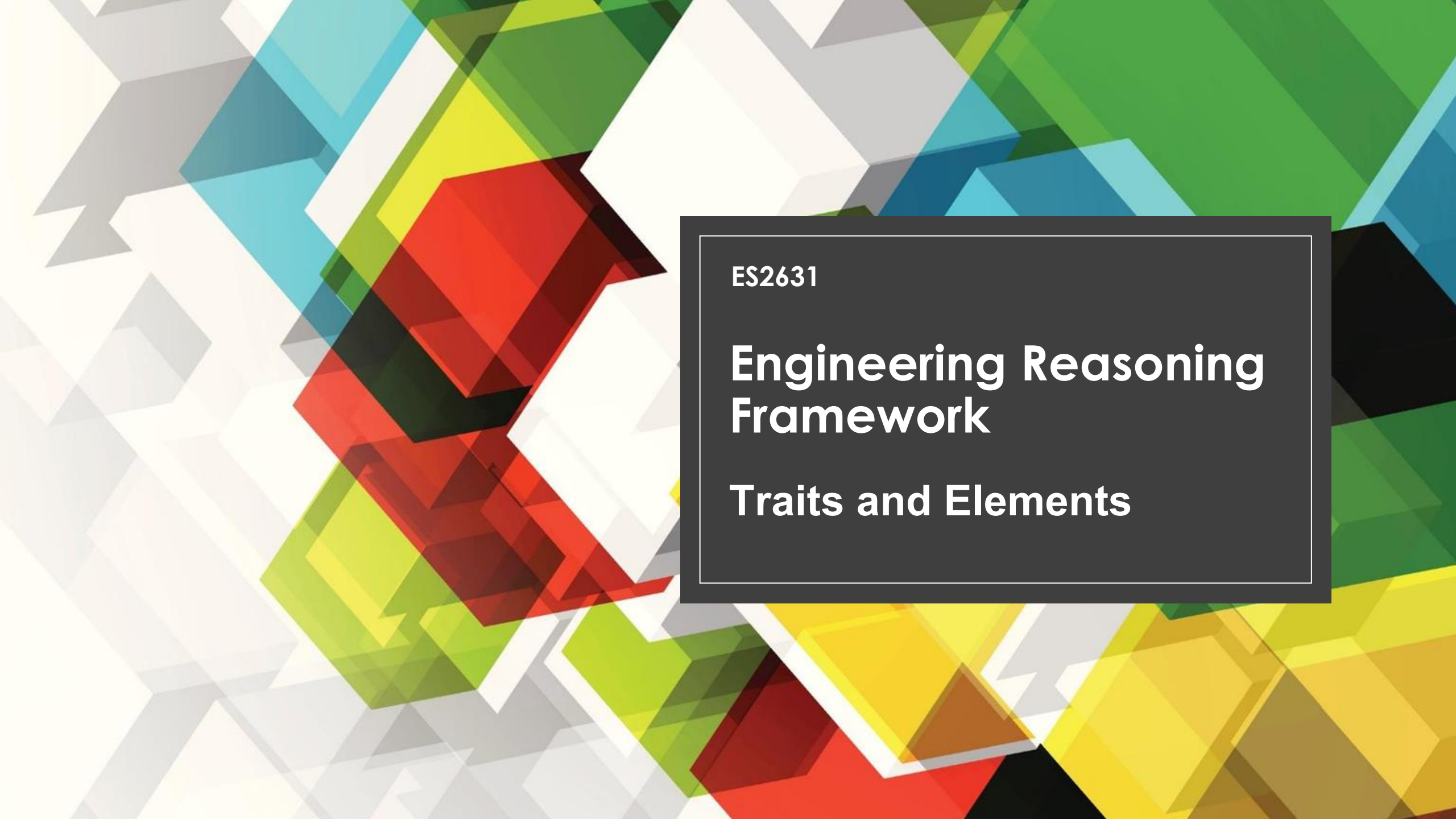




ES2631

Engineering Reasoning Framework

Traits and Elements

Agenda

1. The Framework
2. Intellectual Traits
3. Elements of Thought
4. Application of the Framework





THE FRAMEWORK

Engineering Reasoning Framework

The Problem

Everyone thinks; it is our nature to do so. But much of our thinking, left to itself, is biased, distorted, partial, uninformed or down-right prejudiced. Yet the quality of our life and that of what we produce, make, or build depends precisely on the quality of our thought. Shoddy thinking is costly, both in money and in quality of life. Excellence in thought, however, must be systematically cultivated.

(Paul & Elder, 2006, p.21)

The Solution

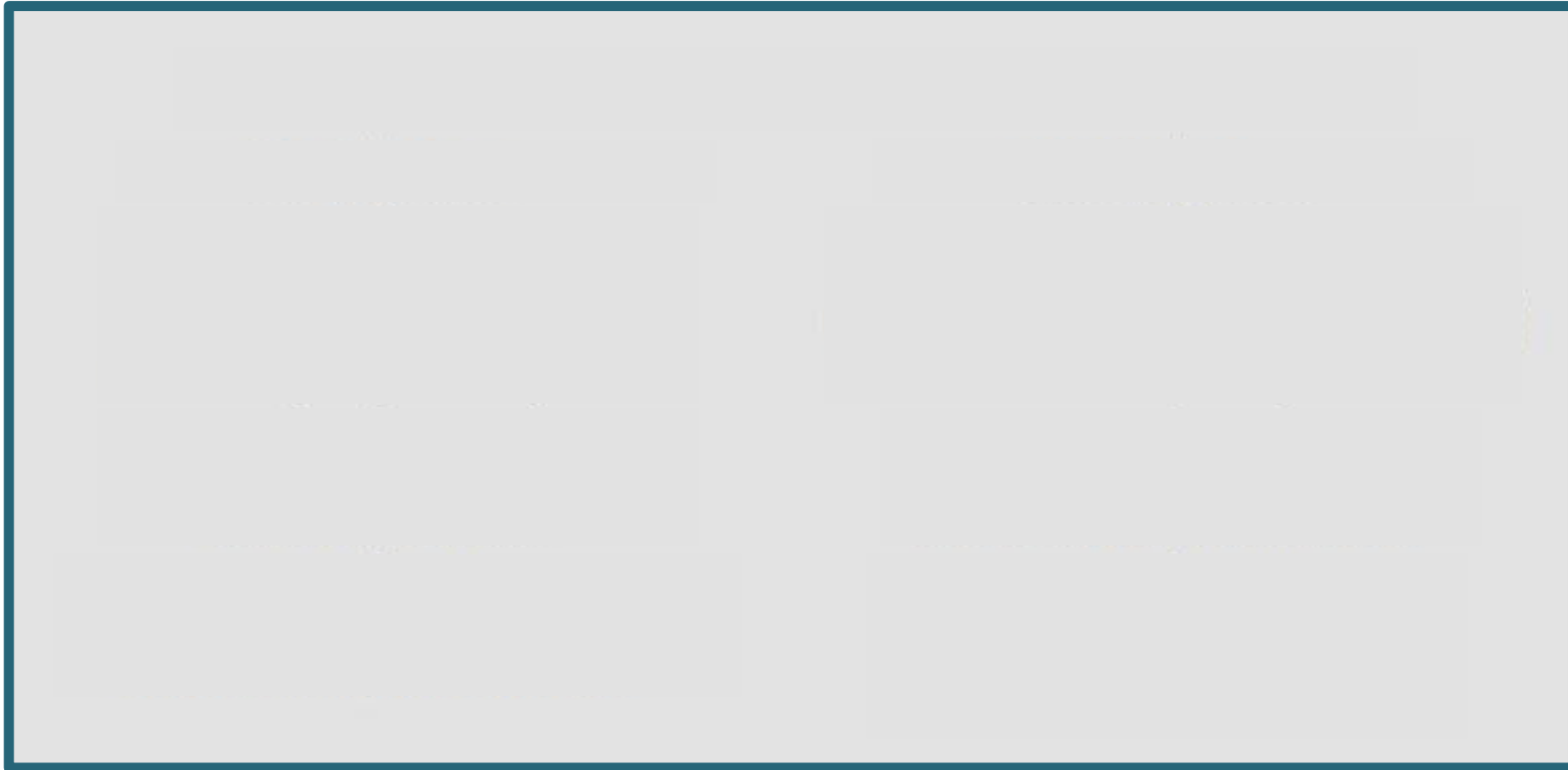
Cultivate critical thought – analyze own and others' thinking

Engineering Reasoning Framework

**“Critical thinking
is the art of
analyzing and evaluating thinking
with a view to improving it.”**

(Paul & Elder, 2006, p. 21)

Two Kinds of Engineering Questions



(Paul et al, 2019, p.16)

Standards are applied to evaluate the quality of a thought or thinking.

THE STANDARDS	
Clarity	Precision
Accuracy	Significance
Relevance	Completeness
Logicalness	Fairness
Breadth	Depth

Must be applied to

Any thought or thinking can be deconstructed by identifying the elements of thought to see what makes up the thought.

THE ELEMENTS	
Purposes	Inferences
Questions	Concepts
Points of view	Implications
Information	Assumptions

As we learn to develop

INTELLECTUAL TRAITS	
Intellectual Humility	Intellectual Perseverance
Intellectual Autonomy	Confidence in Reason
Intellectual Integrity	Intellectual Empathy
Intellectual Courage	Fairmindedness

Constant practice in evaluating thought will result in the development of intellectual traits that define a critical thinker



INTELLECTUAL TRAITS

‘Intellectual traits essential to engineering reasoning’

INTELLECTUAL TRAITS

Intellectual Humility

Intellectual Perseverance

Intellectual Autonomy

Intellectual Empathy

Intellectual Integrity

Confidence in Reason

Intellectual Courage

Fairmindedness

Intellectual Humility

Definition

Being aware of your biases, prejudices, self-deceptive tendencies and the limitations of your viewpoint and experience

QUESTION & STAY OPEN

Questions

- What do I really know about the technological issue I'm facing?
- To what extent do my prejudices, attitudes or experiences bias my judgment?
- Am I quick to admit when I'm in domains beyond my expertise?
- Am I open to consider novel approaches to this problem, and willing to learn and study where warranted?

Intellectual Autonomy

Definition

Thinking for yourself while adhering to standards of rationality

Thinking through issues using your own thinking rather than uncritically accepting the viewpoints, opinions and judgments of others

Questions

- Do I think through technical issues on my own or do I merely accept others' conclusions or judgments?
- Am I willing to stand alone against irrational criticism?

THINK FREELY

Intellectual Integrity

Definition

Holding yourself to the same intellectual standards you expect others to honor (i.e., no double standards)

**BE HONEST & CONSISTENT
IN INTELLECTUAL
STANDARDS**

Questions

To what extent:

- Do I expect of myself what I expect of others?
- Are there contradictions or inconsistencies in the way I deal with technical issues?
- Do I strive to recognize and eliminate self-deception or self-interest when reasoning through engineering issues?

Intellectual Courage

Definition

Disposition to question beliefs that you feel strongly about - includes questioning the beliefs of any culture and sub-culture you belong to

Willingness to express your views even when they are unpopular

SEEK & STAY FIRM IN THE TRUTH

Questions

To what extent:

- have I analyzed the beliefs I hold which may impede my ability to think critically?
- have I demonstrated a willingness to yield my positions when sufficient evidence is presented against them?
- am I willing to stand my ground against the majority (even though others ridicule me)?

Intellectual Perseverance

Definition

Disposition to work your way through intellectual complexities despite the frustration inherent in the task

**DON'T GIVE UP ON
INTELLECTUAL
CHALLENGES**

Questions

- Am I willing to work my way through complexities in an engineering (or design) issue or do I tend to give up when challenged?
- Can I think of a difficult engineering/ design problem in which I have demonstrated patience and tenacity?

Intellectual Empathy

Definition

Awareness of the need to actively entertain views that differ from your own, especially those you disagree with

Entails accurately reconstructing the viewpoints and reasoning of opponents and to reason from premises, assumptions and ideas other than your own

Questions

- Do I listen and seek to understand others' reasoning?
- Do I accurately represent viewpoints I disagree with?
- Do I recognize and appreciate insights in the technical views of others and prejudices in my own?

UNDERSTAND OTHERS' VIEWPOINTS

Confidence in Reason

Definition

Belief that your own higher interests and those of humankind are best served by having the freest play to reason

Using standards of reasonability as the fundamental criteria to judge whether to accept or reject any proposition or position.

**REASON LOGICALLY
BASED ON EVIDENCE**

Questions

- Am I willing to change my position when the evidence leads to a more reasonable position?
- Do I adhere to technical principles and evidence when persuading others of my position or do I distort matters to support my position?
- Do I encourage others to come to their own technical conclusions or do I try to coerce agreement?

Fairmindedness

Definition

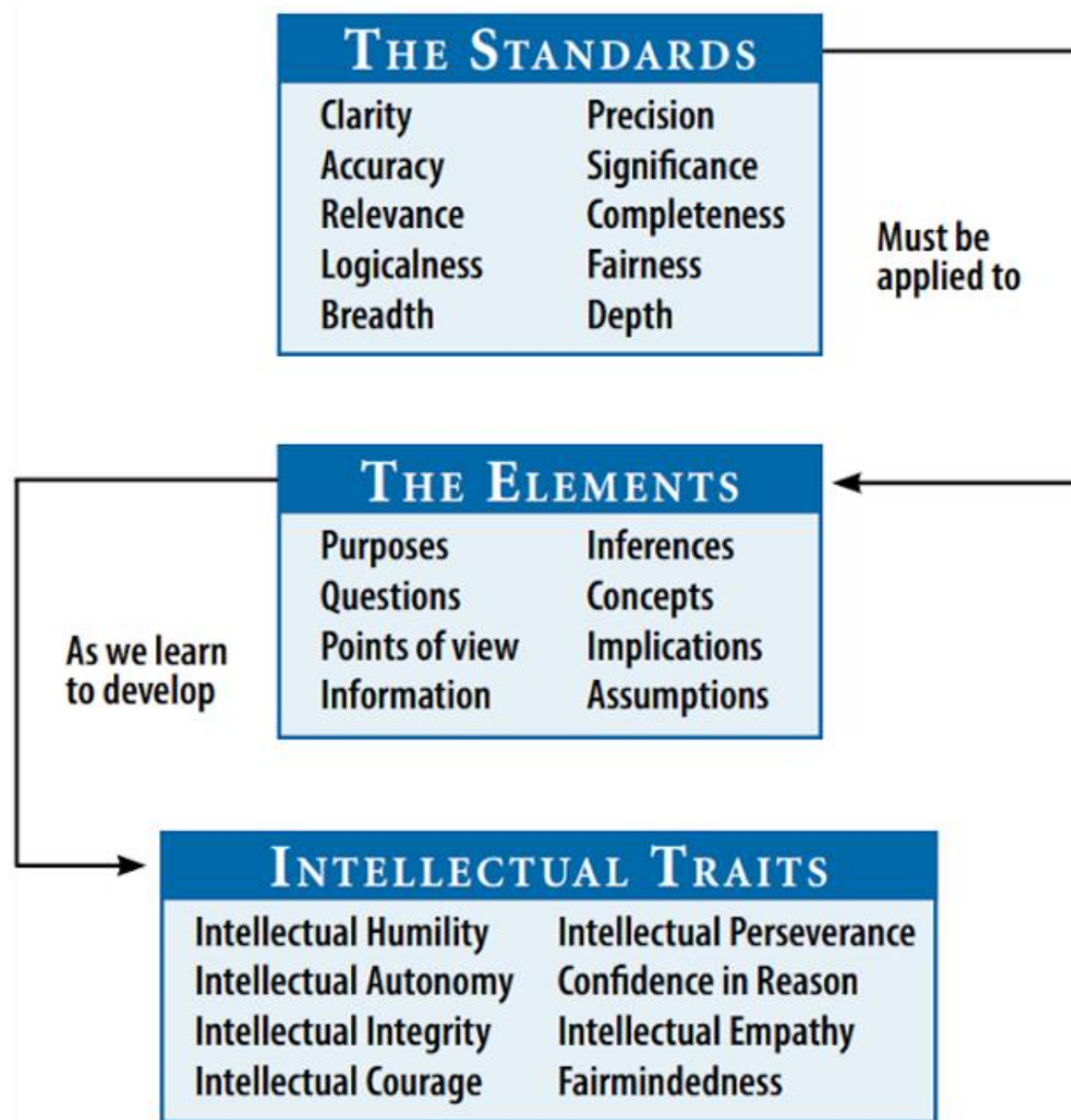
Being conscious of the need to treat all viewpoints alike, without reference to your own feelings or vested interests, or those of your friends, company or nation

Adherence to intellectual standards without reference to your own advantage, or that of your group

Questions

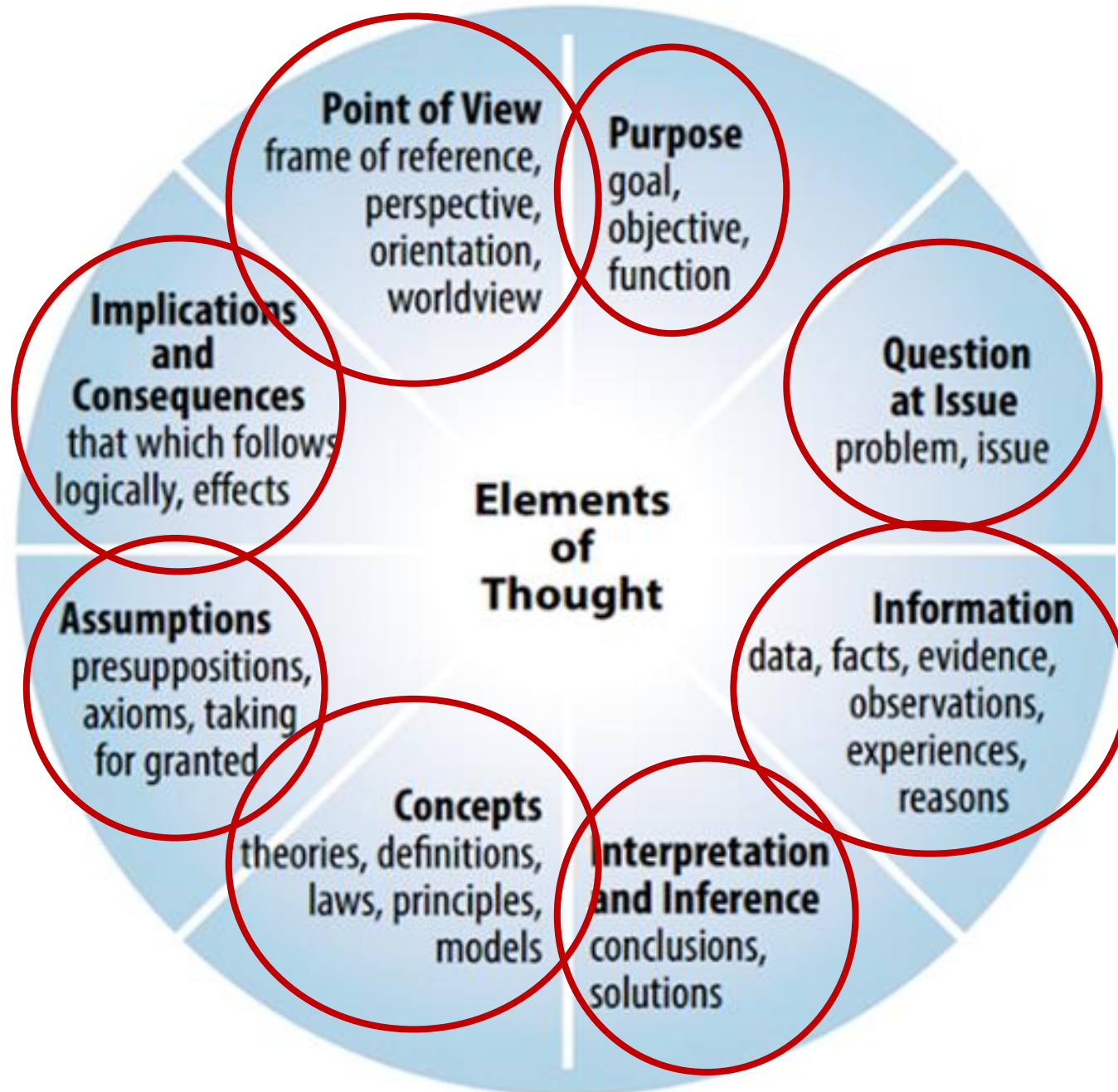
- Am I giving dissenting opinions adequate consideration?
- Has self-interest or bias clouded my judgment?
- Have I adequately thought through the potential hazards to others?

**GIVE EQUAL EMPHASIS TO ALL
VIEWPOINTS/ ARGUMENTS**





ELEMENTS OF THOUGHT



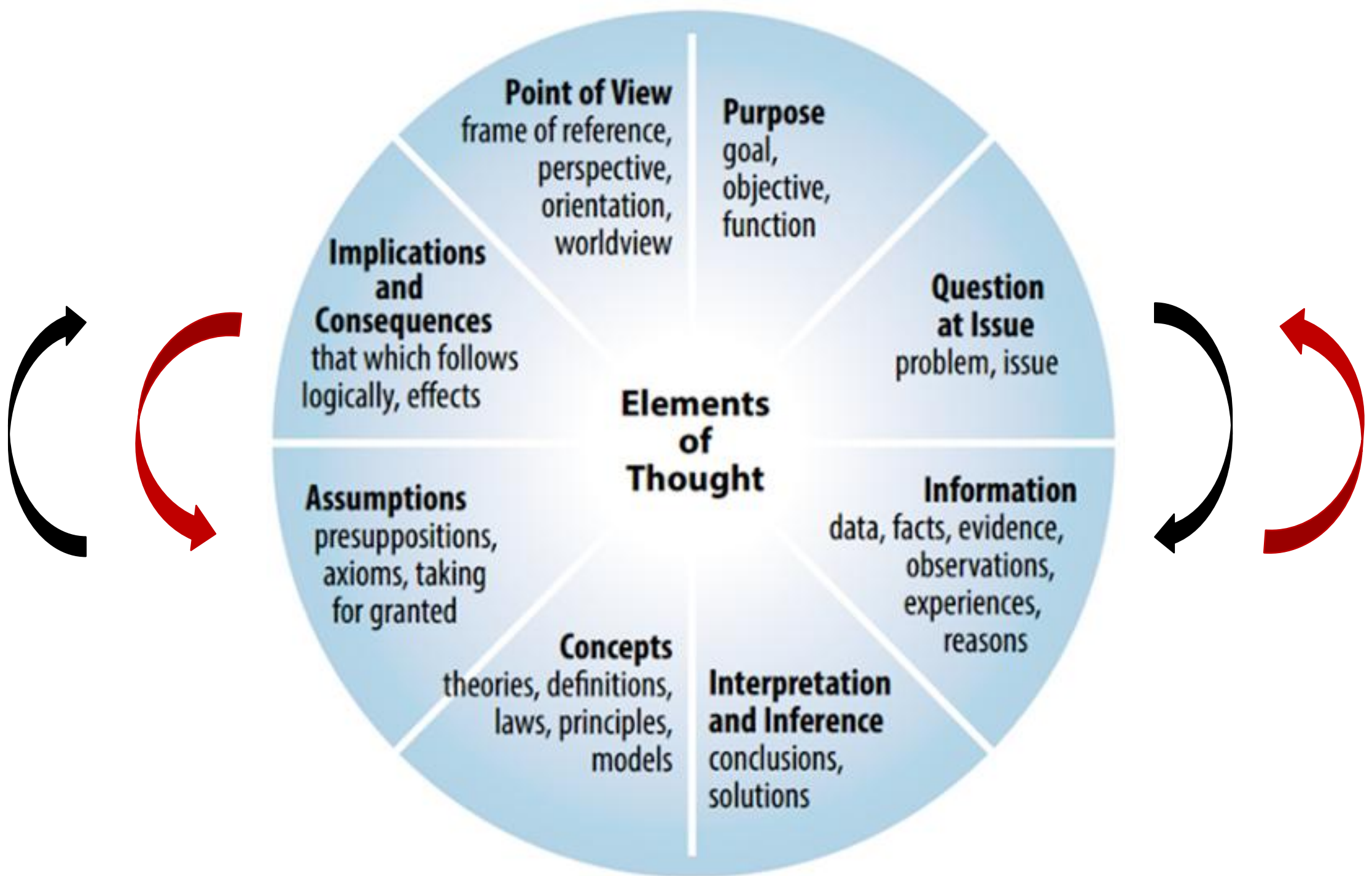
8 Elements of thought

- Purposes
- Questions
- Information
- Concepts
- Inferences
- Assumptions
- Implications
- Point of View

Identifying Elements of Thought

Guide Questions

Purpose	• What is being accomplished? • What is the central aim?
Question	• What question is being raised? • What questions should be addressed? • What complexities in the question are being considered?
Information	• What information is being used in coming to a conclusion? • What information is needed to settle the question? • What shows that the information is true?
Inferences/ Conclusions	• How was this conclusion reached? • Is there another way to interpret the information?
Concepts	• What is the main idea here? • How can this idea be explained?
Assumptions	• What is being for granted? • What assumption has led to this conclusion?
Implications/ Consequences	• If someone accepted this position, what would be some important implications? In other words, what might follow? • What is being implied?
Points of View	• What is the main point of view/ perspective of the issue? • Are there other points of view to consider?



An aerial photograph of a vast, arid desert landscape. The terrain is characterized by rolling hills and deep, eroded canyons. In the lower-left foreground, a small cluster of buildings and a tall, thin tower are visible on a ridge. A dirt road winds through the middle ground. In the distance, a plume of white smoke or steam rises from the valley floor. The sky is hazy, and the overall color palette is dominated by earthy browns and tans.

Yucca Mountain

Nuclear waste repository?

Background

- **1982: U.S. Congress passed Nuclear Waste Policy Act (NWPA) – requires nuclear waste storage repository**
- **1986: Department of Energy (DOE) issued site suitability guidelines.**
- **1987: Yucca Mountain in Nevada was chosen**
- **1994: Tunnelling into Yucca Mountain began**
- **1995: Tunnel boring machines encountered loose ground.**
- **1996: Testing revealed that Yucca Mountain may not be best site to store nuclear waste**
- **Controversy surrounded Yucca Mountain**

Background

- **2012: The Blue Ribbon Commission recommended a new organization to manage spent nuclear fuel.**
- **2012-2016: Controversy continued over Yucca Mountain**
- **2017: Trump administration called for restart of licensing of Yucca Mountain, but later reversed position.**
- **2021: Biden administration opposed Yucca Mountain; announced plans to send waste to places where state, local and tribal governments agree to accept it.**
- **2024: Nuclear waste still stored at nuclear plants across the country**

(Ambrose, 2016; Lupiani, 2018; EPA, 2023)

An aerial photograph of a vast, arid desert landscape. The terrain is characterized by rolling hills and valleys, with a prominent, light-colored, winding path or dry riverbed cutting through the center. The colors range from dark, rocky slopes to lighter, sandy or silty areas. The sky is a pale, hazy blue, suggesting a clear but slightly distant atmosphere. The overall scene conveys a sense of isolation and the rugged beauty of a desert environment.

Yucca Mountain

Why Yucca Mountain was selected

The most accepted solution for the final disposition of high-level nuclear waste is permanent geologic burial:

- **Multiple–barrier solution**
- **Provides second, natural barrier to human-engineered casks, as rocks would contain waste after the containers fail**
- **Locates high-level nuclear waste deep in the earth, so less vulnerable to sabotage and accidents.**

(Eureka County Home, 2019)



Why Yucca Mountain was selected

- In a desert location
- Away from population centers
- Secured 1,000 feet under the surface
- In a closed hydrologic basin
- Protected by natural geologic barriers
- Protected by robust engineered barriers and a flexible design

(Eureka County Home, 2019)

Why Yucca Mountain was opposed

- **In an active seismic region**
- **Presence of numerous earthquake faults and volcanic cinder cones near the site**
- **Presence of pathways that could move groundwater and any escaping radioactive materials rapidly through the site to the aquifer beneath and from there to the accessible environment**
- **Evidence of hydrothermal activity within the proposed repository block**

(Eureka County Home, 2019)

Alternatives to Yucca Mountain

1. **Nanotechnology to separate and neutralize waste elements**
2. **Transmutation to convert waste into less harmful materials**
3. **Genetically engineered microbes to consume the waste elements**
4. **Reprocessing to remove useable fuels and higher-level radionuclides**
5. **High-energy magnetic fields to separate waste components**

BUT

- **concerns over proliferation of nuclear material created.**
- **By-product of reprocessing presents the same problem as nuclear waste – how to dispose of it?**
- **economic concerns**



U.S. DEPARTMENT OF ENERGY
NEVADA TEST SITE
YUCCA MOUNTAIN PROJECT



Yucca Mountain as nuclear waste repository



Photograph by Associated Press

(Source: Wald, 2011)



Photograph by Iris Dumuk

(Source: Las Vegas Weekly, 2008)

Engineering Reasoning Framework

Elements

Standards

Yucca Mountain

- **Question at issue** (question to answer or problem to solve)
 - * Yucca Mountain as a nuclear waste repository: Yes or No?
- **Purpose** (aim and objective of the thinking)
 - * To determine if Yucca Mountain is suitable for storing nuclear waste.

Yucca Mountain

- Concepts

- * Site characterization (aka site investigation/assessment)
- * Sacredness of the site

- Information

- * The location of the proposed repository is a block of rock 1,000 m beneath the surface and 800 m above the water table below.
- * The mountain receives six to seven inches of rain and snow per year.

(Eureka County Home, 2019)

Yucca Mountain

- **Inferences** (conclusions drawn from information)
 - * The site is stable
 - * The chance of the radioactive waste leaking from the mountain is small.
- **Implications** (consequences and outcomes)

Yucca Mountain used as repository →

 - * economic returns for the communities nearby
 - * possible contamination of the groundwater

Yucca Mountain

- **Assumptions** (points or ideas taken for granted)
 - * Yucca Mountain will not erupt for millions of years.
- **Points of view** (ways of looking at it)
 - * Yucca Mountain is safe for storing radioactive waste.
 - * Yucca Mountain is not safe for storing nuclear waste.
 - * More could be done to make Yucca Mountain a safe repository for nuclear waste.

Yucca Mountain





APPLICATION in Assignments

Application of Framework

1. Assignment 1
 - Identify a problem
 - Design an engineering solution
2. Assignment 2
 - Critique an engineering conceptual design
3. Assignment 3
 - Critique an article

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